torus. One complication of this representation is that the doughnut no longer looks flat! However, it actually is. Using the notion of curvature given in the previous endnote, you'd find that all triangles drawn on the surface of the doughnut have angles that add up to 180 degrees. The fact that the doughnut looks curved is an artifact of how we've embedded a two-dimensional shape in our three-dimensional world. For this reason, in the current context it is more useful to use the manifestly uncurved representations of the two- and three-dimensional tori, as discussed in the text.

13. Notice that we've been loose in distinguishing the concepts of shape and curvature. There are three types of *curvatures* for completely symmetric space: positive, zero, and negative. But two shapes can have the same curvature and yet not be identical, with the simplest example being the flat video screen and the flat infinite tabletop. Thus, symmetry allows us to narrow down the curvature of space to three possibilities, but there are somewhat more than three shapes for space (differing in what mathematicians call their global properties) that realize these three curvatures.

14. So far, we've focused exclusively on the curvature of three-dimensional space— the curvature of the spatial slices in the spacetime loaf. However, although it's hard to picture, in all three cases of spatial curvature (positive, zero, negative), the whole four-dimensional spacetime is curved, with the degree of curvature becoming ever larger as we examine the universe ever closer to the big bang. In fact, near the moment of the big bang, the four-dimensional curvature of spacetime grows so large that Einstein's equations break down. We will discuss this further in later chapters.

Chapter 9

1. If you raised the temperature much higher, you'd find a fourth state of matter known as a *plasma*, in which atoms disintegrate into their component particles.

2. There are curious substances, such as Rochelle salts, which become less ordered at high temperatures, and more ordered at low temperatures —the reverse of what we normally expect.